



COMPLIANCE CHECK FRAMEWORK FOR UNIVERSITY BUSINESS PROCESSES

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Abstract

Higher Education Institutions (HEIs) operate within a complex ecosystem of national laws, such as the Council of Higher Education (YÖK) regulations, and internal institutional directives. Ensuring alignment between these heterogeneous document sets is traditionally a manual, labor-intensive, and error-prone process. This project proposes a comprehensive "Compliance Check Framework" designed to automate the semantic verification of university business processes against regulatory standards. Utilizing Natural Language Processing (NLP) and Large Language Models (LLMs), the framework establishes a robust pipeline for document preprocessing, embedding generation, and semantic retrieval.

The core contribution of this study is a comparative benchmarking of diverse Retrieval-Augmented Generation (RAG) architectures to identify the most effective methodology for the legal-educational domain. The system evaluates and contrasts Vanilla, Sparse, Dense, Hybrid, Multi-query, Multi-hop, and Graph-RAG approaches. By integrating Chain-of-Thought (CoT) reasoning, the framework not only calculates a numerical compliance score (0–100) but also generates interpretable, evidence-based explanations for non-compliant cases. Preliminary designs indicate that while dense retrieval captures semantic nuance, hybrid and graph-based approaches significantly enhance the detection of complex regulatory dependencies. This system aims to provide university administrators with a scalable, transparent, and automated decision-support tool.

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LIST OF ABBREVIATIONS

RAG	Retrieval-Augmented Generation
NLP	Natural Language Processing
LLM	Large Language Model
YÖK	Yükseköğretim Kurulu (Council of Higher Education)
CoT	Chain-of-Thought
SBERT	Sentence Bidirectional Encoder Representations from Transformers

1 Introduction

In the modern landscape of higher education, universities operate within a strict legal framework. Every administrative and academic decision—from student grading systems to graduation requirements and disciplinary actions—must comply with national laws. In Turkey, the primary legal authority is the **Higher Education Law No. 2547 (YÖK Law)**. Based on this central law, institutions like Istanbul Bilgi University create their own internal directives, such as the "Undergraduate Education and Examination Regulation". However, ensuring that these internal university documents are always perfectly aligned with the complex and frequently changing national laws is a major challenge. Currently, this compliance checking process is performed manually by administrative staff. This manual method is slow, difficult to scale, and highly prone to human error. A simple oversight in a regulation can lead to legal disputes or loss of student rights. Therefore, there is an urgent need for automated systems that can perform this verification faster and more accurately using Artificial Intelligence (AI).

Recent advancements in **Natural Language Processing (NLP)** and **Large Language Models (LLMs)** offer powerful solutions to this problem. Specifically, a technique called **Retrieval-Augmented Generation (RAG)** has become the standard for grounding AI responses in factual documents. Studies in the financial and legal domains have successfully used RAG for compliance. For example, Sun et al. (2025)[1] proposed a three-layered RAG framework to check business processes, while Masoudifard et al. (2024)[2] demonstrated how **Graph-RAG** can improve the traceability of requirements by modeling relationships between documents. Despite these advancements, most existing models are designed for finance or general law. There is a significant gap in the literature for frameworks specifically optimized for the hierarchical and text-heavy domain of **Higher Education**.

Proposed Solution

This project addresses this gap by developing a specialized "Compliance Check Framework" that automates the semantic verification of university business processes against YÖK regulations. Unlike standard systems that rely on a single search method, this study adopts a "Multi-Strategy" approach. We research and integrate a wide variety of advanced RAG architectures to find the best possible solution.

To ensure precision at every stage, our framework utilizes specific models and techniques:

- 1) **Data Preprocessing and Embedding:** First, we convert legal

PDFs into text. To represent these texts mathematically, we use the **SBERT (Sentence-BERT)** model. SBERT allows us to capture the semantic meaning of paragraphs, ensuring the system understands the context rather than just matching words.

2) Retrieval Strategy (The "Omni" Approach): We implement an agent-based architecture (Omni-RAG) that dynamically selects the best retrieval method:

- **Sparse Retrieval (BM25):** Used to find exact matches for specific article numbers (e.g., "Article 44") or unique legal terms.
- **Dense Retrieval:** Uses vector databases to find semantically similar content even if different words are used.
- **Hybrid RAG:** Combines Sparse and Dense methods to capture both keywords and meaning, as suggested by Hillebrand et al. (2024)[4].
- **Graph-RAG:** Since laws often reference each other, we model regulations as a knowledge graph. This allows the system to see structural relationships that standard methods miss.
- **Multi-hop and Multi-query:** For complex questions requiring information from multiple documents, the system breaks the query into parts and "hops" between related legal articles to build a complete answer.

3) Reasoning and Reporting: For the final assessment, we employ advanced LLMs such as OpenAI GPT-4o or LLaMA-2. We use Chain-of-Thought (CoT) prompting to force the model to explain its reasoning step-by-step. The system then calculates a numerical compliance score (0–100) and generates a transparent report.

Differentiation from Related Works

This study differs from existing works in three key aspects. First, while studies like Li et al. (2025)[3] focus on financial regulations and converting compliance into code, our work is the first to apply these advanced RAG techniques specifically to the Turkish Higher Education context. Second, we do not rely on a single retrieval method; instead, we perform a comparative benchmark of Vanilla RAG, Graph-RAG, and Prompt-Enhanced RAG within the same framework to empirically determine which is best for legal texts. Third, our system prioritizes explainability; it does not just say "Non-Compliant" but provides evidence-based reasoning citing specific YÖK articles.

Main Contributions

The main contributions of this study are summarized as follows:

1) Development of a Domain-Specific Framework: We propose the first comprehensive compliance checking framework tailored specifically for the hierarchical structure of Higher Education regulations (YÖK Laws vs. University Directives).

2) Comparative Analysis of RAG Architectures: We provide a benchmark comparison of diverse RAG approaches—including Vanilla, Sparse, Dense, Hybrid, and Graph-RAG—to identify the most accurate method for detecting legal inconsistencies.

3) Explainable AI for Auditing: By integrating Chain-of-Thought (CoT) reasoning, the system not only provides a numerical compliance score but also generates transparent, human-readable explanations for why a specific process is compliant or non-compliant.

4) Digitization of University Governance: The proposed Information Portal and automated reporting tools reduce the manual workload of administrative staff, minimizing human error and ensuring continuous alignment with changing national laws.

2 Related Works

The automation of regulatory compliance checking has evolved significantly in recent years. Historically, this process relied on manual labor, which is slow and prone to errors. With the rise of Artificial Intelligence (AI), the field has shifted from basic text analysis to advanced **Retrieval-Augmented Generation (RAG)** architectures. This section reviews existing literature, categorizing methodologies into retrieval strategies, reasoning capabilities, and domain-specific applications. We specifically analyze variants such as **Vanilla, Sparse, Dense, Hybrid, Graph, and Omni-RAG** to position our framework within the current state of the art.

Early NLP Approaches in Legal Automation

Before the widespread adoption of Large Language Models (LLMs), compliance automation relied on traditional Natural Language Processing (NLP). Singh (2024)[11] demonstrated the use of **Topic Modeling (LDA)** to categorize legal documents and extract key insights from unstructured text. Similarly, Nithya et al. (2024)[6] utilized NLP pipelines involving **Named Entity Recognition (NER)** and summarization to streamline the analysis of legal contracts and court judgments. Abualhaija et al. (2022)[5] proposed automated **Question Answering (QA)** systems using BERT-based models to extract requirements from regulations. While these methods were effective for identifying keywords and classifying documents, they lacked the deep reasoning capabilities required to understand the complex relationships between different laws.

Retrieval Architectures: From Vanilla to Hybrid RAG

To overcome the limitations of traditional NLP, researchers adopted RAG, which connects generative AI with factual document databases.

Vanilla and Dense RAG: The standard approach, often called "Vanilla RAG", relies on Dense Retrieval. This method converts text into mathematical vectors (embeddings) to find semantic similarities. Sun et al. (2025)[1] successfully applied this in a three-layer framework for business process compliance, using SBERT to map regulatory texts into computable vectors.

Sparse and Hybrid RAG: While Dense retrieval is powerful for understanding context, it can miss exact keyword matches, such as specific article numbers. To address this, Hillebrand et al. (2024)[4] introduced a Hybrid

RAG chatbot for risk assurance. Their study showed that combining Sparse Retrieval (keyword-based algorithms like BM25) with Dense vectors significantly improves accuracy by capturing both precise terminology and general meaning.

Structural and Relational Reasoning: Graph and Multi-hop RAG

Standard vector searches often treat text chunks in isolation, missing the hierarchical connections between laws.

Graph-RAG: To capture the interconnected nature of regulations, Masoudifard et al. (2024)[2] proposed Graph-RAG. By modeling regulations as a Knowledge Graph (nodes and edges), their system preserved the structural links between requirements, outperforming baseline methods in traceability tasks. Similarly, Sun et al. (2025)[1] utilized an "eventic graph" to structure regulatory information around actions and states rather than just keywords

Multi-hop and Multi-query RAG: Complex compliance questions often require information from multiple disconnected documents. Masoudifard et al. (2024)[2] demonstrated that integrating Graph-RAG with Chain-of-Thought (CoT) prompting enables Multi-hop reasoning. This allows the model to "hop" from one regulation to another to build a complete answer. Additionally, Multi-query strategies decompose complex user questions into smaller sub-queries to retrieve a broader range of relevant information.

Logic and Code Generation

Beyond retrieving text, recent studies have explored translating regulations into executable logic. Li et al. (2025)[3] introduced the "Compliance-to-Code" framework. Instead of just analyzing text, their model converts financial regulations into executable Python code modules, enabling deterministic verification of compliance rules. Fazlija et al. (2025)[9] also benchmarked LLMs for generating code from financial regulations (Basel III), highlighting the potential of AI to automate the implementation of complex standards.

Research Gap and Contribution

Despite these advancements, a systematic literature review by Zadenoori et al. (2025)[10] notes that most LLM applications in Requirements Engineering focus on generation rather than rigorous compliance checking in

specific domains. Existing frameworks are predominantly tailored for the financial (Li et al.[3], 2025; Fazlija et al., 2025[9]) or general legal sectors. There is a significant lack of frameworks specifically designed for Higher Education Institutions, where internal directives must align with national laws (e.g., YÖK Law No. 2547) in a hierarchical manner.

This project addresses this gap by proposing a Multi-Strategy (Omni) framework. Unlike previous studies that rely on a single retrieval method, we benchmark Vanilla, Sparse, Dense, Hybrid, and Graph-RAG architectures specifically for the Turkish educational domain. By integrating Chain-of-Thought (CoT) reasoning, our system not only detects violations but also provides explainable, evidence-based reports.

3 Tables

3.1 Table 1: Summary of Related Works

Author / Year	Task / Focus	Model / Technique	Dataset / Domain	Performance / Contribution
Sun et al. (2025)[1]	Automated regulatory compliance checking of institutional and corporate processes	Three-layer RAG architecture (Static, Dynamic, Computational) integrating SBERT, LLaMA-2, and event-based graphs	Legal & institutional documents (EU2UK, GDPR-13, CONTRACT, CSSCD)	Achieved state-of-the-art results with accuracy above 90%; demonstrated global reasoning and conflict detection across regulations
Masoudifard et al. (2024)[2]	Requirement traceability and compliance reasoning in regulated environments	Graph-RAG combined with Chain-of-Thought (CoT) and Tree-of-Thought (ToT) prompting	Multi-domain legal text (NASA X-38 Fault Tolerant System Services, Iranian Online Broker SRS)	Graph-RAG with ToT achieved highest F1-scores; improved contextual retrieval and multi-hop reasoning
Li et al. (2025)[3]	Code-based compliance automation via regulation-to-code transformation	FinCheck pipeline using SFT fine-tuning with GLM-4 and Qwen-72B / Qwen3-8B	Financial regulations (Chinese dataset from Beijing Stock Exchange)	GLM-4 excelled in structural reasoning; Qwen achieved up to 78.1% pass@1 in code generation

Fazlija et al. (2024/2025)[9]	Financial regulation implementation and benchmarking of text-to-code approaches	Zero-shot, Few-shot, and Dynamic Few-shot learning with multiple LLMs	Basel III Standardised Approach for credit risk; over 6,000 test cases	Achieved pass@1 up to 75.38% and pass@10 up to 91.67%; proposed human-AI collaborative workflows
Abualhaija et al. (2022)[5]	Automated question answering for compliance and requirements engineering	Two-phase approach: IR-based retrieval (BCE similarity) + MRC using BERT / RoBERTa	Multi-document compliance datasets (GDPR, EU Directives 2019/770, 2019/771)	Retrieved correct answers in top-5 passages in 94% of cases; RoBERTa achieved 91% accuracy
Ahmad et al. (2025)[9]	Financial regulation compliance analysis and report generation	RAG combined with LLM-based reasoning	Corporate policy documents	Developed a compliance scoring system (0–100) with automated, interpretable reports
Hillebrand et al. (2024)[4]	Risk and quality compliance chatbot development	Hybrid retrieval (BM25 + Dense embeddings)	PwC industrial compliance dataset	Deployed industrial-scale compliance chatbot; performance evaluated using DeepEval framework
Nithya et al. (2024)[6]	Legal process automation including drafting, validation, and classification	NLP techniques (NER, summarization) using Transformers (BERT / GPT)	General legal datasets and case laws (Indian legal sector)	Automated clause extraction; improved efficiency in legal research and document handling

Singh et al. (2024)[11]	Legal text topic modeling and summarization	Latent Dirichlet Allocation (LDA) and Transformer-based summarization	Legal text corpus from Indian judiciary (contracts, NDAs, court acts)	Successfully identified latent topics and generated concise summaries for legal practitioners
Kaur et al. (2024)[10]	Requirements validation using large language models	GPT and LLaMA-based models	Requirements engineering corpora	Comprehensive systematic review highlighting strengths and limitations of LLMs in RE tasks
Shinde et al. (2025)[7]	Legal document classification and entity extraction	Legal-BERT, CNN, and LSTM architectures	Public legal text datasets	Established baseline results for legal entity extraction and document summarization

Table 1: Summary of Related Works on RAG-Based Compliance and Legal Automation

Table 1 provides a comparative overview of recent studies that apply Retrieval-Augmented Generation (RAG), graph-based reasoning, and large language models to regulatory compliance, requirement traceability, and legal automation tasks. The table highlights differences in application domains, ranging from financial regulations to general legal and requirements engineering datasets. It also illustrates that most existing works focus on corporate or financial compliance, while none of them directly address the hierarchical structure of higher education regulations. This observation further motivates the need for a domain-specific compliance framework tailored to YÖK laws and university directives.

3.2 Table 2: Summary of Datasets and Domains

Domain	Dataset / Source	Data Type	Size / Instances	Used in
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Legal regulations	EU / Turkish Law PDFs	Law text corpus	~5,000 clauses	Sun et al. (2025)[1]
Corporate compliance	PwC Regulation Set	Policy documents	2,000+ docs	Hillebrand et al. (2024)[4]
Financial sector	Basel / IFRS / Bank Regulations	Legal + procedural text	1,500 rules	Ahmad et al. (2025)[9], Li et al. (2025)[3]
Requirements analysis	Software Req. Corpus (NASA / RE'24)	Technical text	800+ requirements	Kaur et al. (2024)[10]
General legal text	Legal NLP Datasets (LDA, MURA-like)	Legal text (cases & acts)	10k+ docs	Singh et al. (2024)[11], Nithya et al. (2024)[6]

Table 2: Summary of Datasets and Application Domains Used in Related Works

Table 2 summarizes the datasets and application domains employed in existing compliance and legal automation studies. The majority of datasets are derived from financial regulations, corporate policies, and general legal corpora. Notably, there is a lack of publicly available datasets specifically designed for higher education governance and national education laws such as YÖK Law No. 2547. This gap indicates that the proposed framework operates in a relatively unexplored domain and contributes novel experimental insights for compliance checking in higher education institutions.

4 System Design

This section presents the architectural blueprint of the proposed Compliance Check Framework. The system is designed not merely as a text comparison tool, but as a comprehensive, multi-layered *Omni-RAG* platform. As illustrated in the high-level architecture diagram Figure 1, the framework integrates multiple retrieval strategies and reasoning engines to handle the complex and hierarchical nature of higher education regulations.

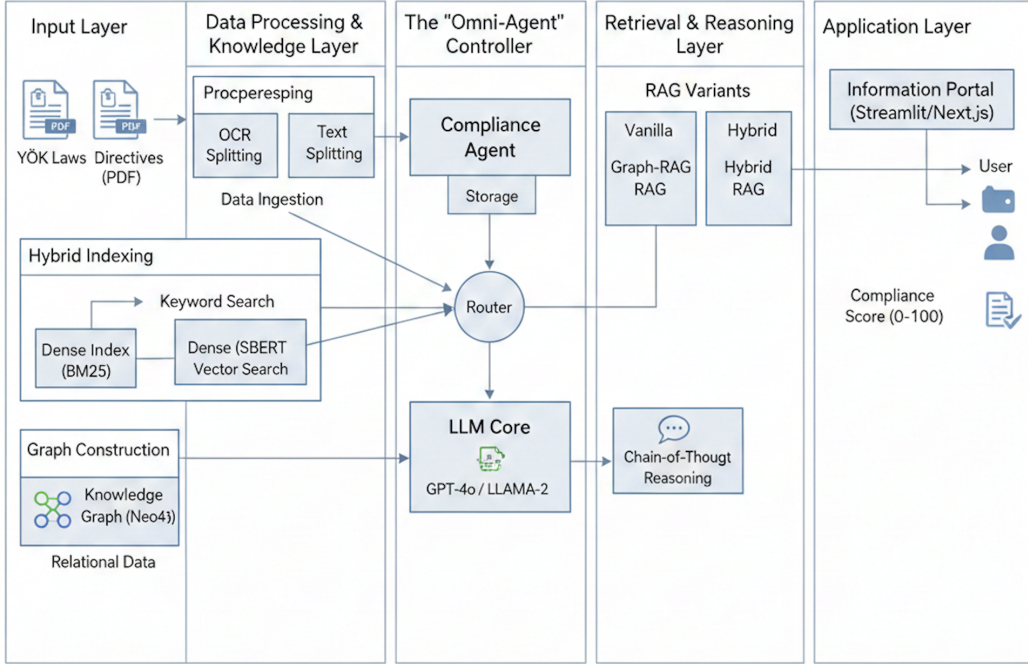


Figure 1: High-Level System Architecture Diagram

4.1 High-Level Architecture: The *Omni-Agent* Approach

The core innovation of our design is the *Omni-Agent Controller* (Source 824). Unlike traditional pipelines that follow a fixed and linear sequence, our framework employs an intelligent agent that acts as the central decision-making component. The agent receives the user query and dynamically selects the most appropriate tool, retrieval method, and reasoning path to produce an accurate and explainable output. This modular design enables us to benchmark and combine different RAG variants (e.g., Vanilla, Hybrid, and Graph-based approaches) under a single unified environment.

The architecture is structured into four primary layers:

1. **Input Layer:** Handles raw PDF documents such as YÖK laws, regulations, and internal university directives.
2. **Data Processing & Knowledge Layer:** Transforms raw text into machine-readable representations and builds searchable indexes.
3. **Retrieval & Reasoning Layer:** Executes semantic search and compliance reasoning using AI models.
4. **Application Layer:** Provides the user interface for interaction, reporting, and transparency.

4.2 Data Processing and Knowledge Layer

The framework relies on high-quality document ingestion, which directly affects retrieval accuracy and downstream reasoning. As shown in the system workflow 1, this layer consists of the following steps:

- **Preprocessing:** PDF documents are converted into clean text through OCR (Optical Character Recognition) when necessary, followed by normalization and semantic chunking. Chunking is applied to produce coherent segments that are suitable for both sparse and dense retrieval.
- **Hybrid Indexing:** To improve robustness across different query styles, the corpus is indexed in two complementary ways:
 - **Sparse Index (BM25):** Supports exact keyword matching and is critical for locating references such as explicit article numbers and domain-specific terminology.
 - **Dense Index (SBERT):** Sentence-BERT embeddings are used to represent each chunk in a vector space. This enables semantic retrieval even when the user query and legal text use different wording.
- **Graph Construction:** In parallel, a Knowledge Graph is built (e.g., using Neo4j) to model relationships between legal entities, including references, dependencies, and hierarchical structure. This representation is essential for Graph-RAG operations, where answering a query may require navigating across linked regulations.

4.3 Retrieval and Reasoning Engine

This layer is the core execution engine where the *Omni-Agent* demonstrates its routing capabilities. The Router component directs each query to the most appropriate retrieval strategy:

- **Vanilla RAG:** Uses dense vector retrieval for general context and semantic similarity queries.
- **Hybrid RAG:** Combines results from sparse (BM25) and dense (SBERT) retrieval to maximize precision and recall, particularly for mixed queries involving both keywords and semantics.
- **Graph-RAG:** Traverses the Knowledge Graph to answer queries that require multi-hop reasoning across linked legal texts (e.g., when one law clause refers to another).

After retrieval, the selected evidence passages are forwarded to the LLM Core (powered by models such as GPT-4o or LLaMA-2). The model applies Chain-of-Thought (CoT) reasoning to analyze the retrieved evidence step-by-step. In compliance checking mode, the LLM compares the university’s internal process statements against the retrieved YÖK clauses to identify compliance, partial compliance, or non-compliance.

4.4 Application Layer

The system outputs are delivered through an interactive Information Portal implemented using Streamlit or a web stack such as Next.js. The interface is designed for transparency and practical usage by university staff. Specifically, the portal enables users to:

1. Upload and manage regulation documents and internal directives,
2. View the calculated Compliance Score (0–100),
3. Read a detailed explanation describing why a document is compliant or non-compliant, supported by retrieved legal evidence.

4.5 System Workflow (Sequence of Operations)

The operational flow of the framework, from data ingestion to final reporting, follows a strict sequence as depicted in the project flowchart:

1. **Data Collection:** The system gathers YÖK laws and university directives.
2. **Preprocessing:** Documents are cleaned, normalized, and split into semantic chunks.
3. **Embedding Generation:** SBERT converts text chunks into numerical vector representations (Source 884).
4. **RAG Comparison:** The *Omni-Agent* retrieves relevant legal evidence using Vanilla, Hybrid, or Graph-based search strategies.
5. **Reasoning (CoT):** The LLM evaluates alignment and detects compliance gaps through step-by-step reasoning.
6. **Reporting:** The system generates a compliance report and an aggregated compliance score for decision support.

4.6 User Interface Design and Interaction Flow

To facilitate user interaction and ensure transparency, we designed a web-based Information Portal. Figure 2 illustrates the proposed dashboard interface, which serves as the primary touchpoint for university administrators.

The interface is designed with three core functional areas:

1. **Compliance Score Visualization:** As shown in the dashboard, the system calculates an aggregate *Total Compliance Score* derived from the RAG-based evaluation logic 1. A visual status badge such as “*Partially Compliant*” provides an immediate, high-level assessment of the uploaded document. This component enables administrators to quickly interpret the overall regulatory alignment without analyzing detailed textual explanations.
2. **Document Management:** The *File Upload* section allows staff to upload internal regulation documents in PDF format, which initiates the automated preprocessing, chunking, embedding generation, and indexing pipeline described in the methodology 1. This design ensures that the compliance analysis process is seamlessly integrated into existing administrative workflows with minimal technical overhead.
3. **Real-Time Alerts:** The *Recent Alerts* panel highlights critical discrepancies detected by the reasoning engine. For instance, the system can flag conflicts such as “*Article 15 contradicts YÖK Law No.*”

2547", enabling administrators to immediately identify and prioritize non-compliance issues 2. These alerts support proactive regulatory management by drawing attention to the most critical inconsistencies.

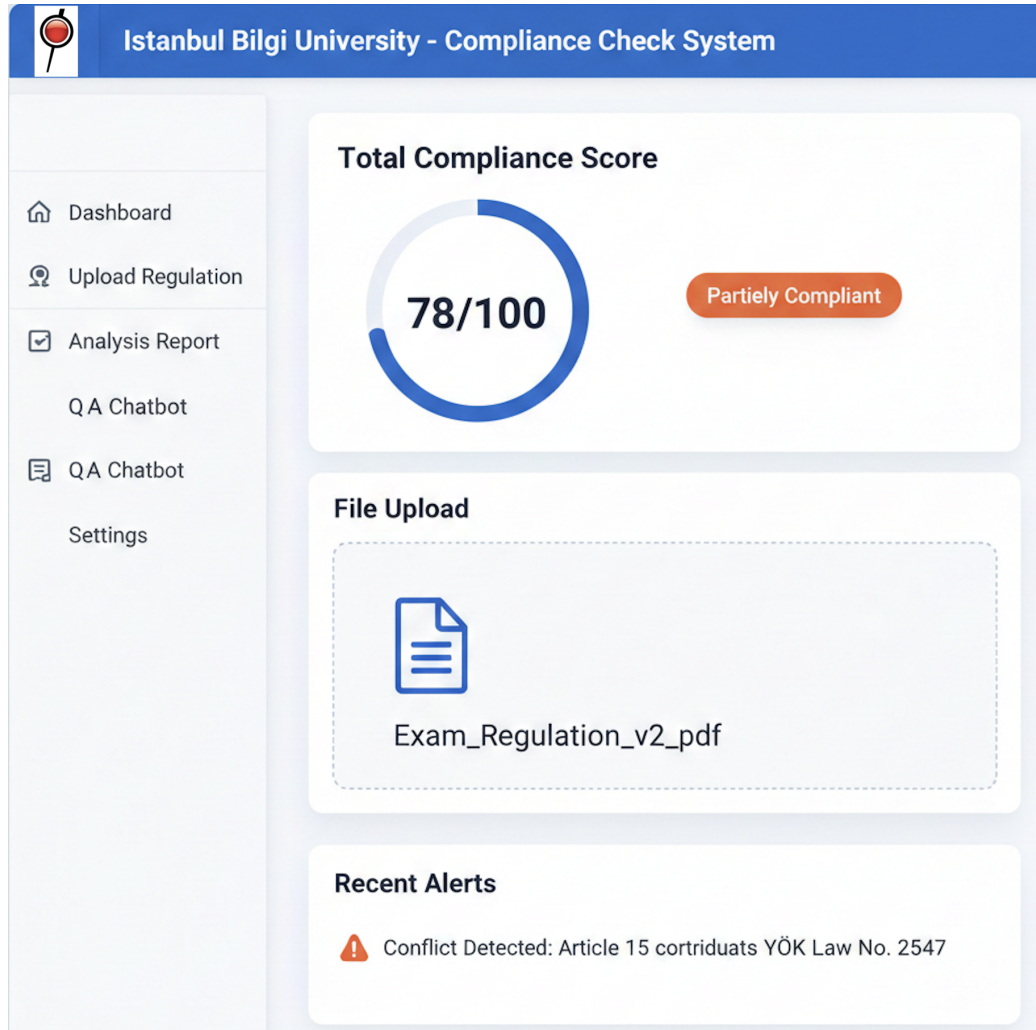


Figure 2: Proposed User Interface (UI) for the Information Portal. The dashboard allows administrators to upload documents, visualize the calculated Compliance Score, and review specific alerts regarding conflicts between University Directives and YÖK Laws.

5 Methodology

This study follows a system-oriented and experimental methodology to design, implement, and evaluate an automated compliance checking framework for higher education regulations. The methodology is structured to ensure that each component of the proposed *Omni-RAG* architecture is systematically developed, tested, and compared. Rather than treating compliance checking as a single retrieval task, we model it as a multi-stage reasoning pipeline that integrates document processing, semantic retrieval, graph-based reasoning, and Large Language Model (LLM) explainability.

The overall workflow of the methodology is illustrated through four main phases:

1. Data collection and preprocessing,
2. Knowledge representation and indexing,
3. Multi-strategy RAG-based compliance evaluation,
4. Explainable reporting and experimental benchmarking.

This structured approach ensures consistency with the system design presented in Section 6 and enables reproducible evaluation of different RAG architectures under the same experimental environment.

5.1 Phase 1: Data Collection and Preprocessing

Two primary document corpora are constructed:

- **Legal Corpus:** National regulations and YÖK laws, including Higher Education Law No. 2547 and related directives.
- **Institutional Corpus:** Internal university regulations, examination procedures, and workflow documents.

All documents are collected in PDF format and converted into machine-readable text using OCR techniques when necessary. The preprocessing pipeline consists of:

- text normalization and cleaning,
- sentence segmentation and paragraph detection,
- removal of irrelevant formatting artifacts,

- semantic chunking of documents into coherent textual blocks.

Chunking is applied to preserve contextual integrity while maintaining compatibility with embedding models and retrieval systems. Each chunk represents a meaningful regulatory or procedural statement that can independently participate in compliance evaluation.

5.2 Phase 2: Knowledge Representation and Index Construction

To support different retrieval strategies, the knowledge base is indexed in three complementary structures:

- **Sparse Index (BM25):** Enables exact keyword matching, especially useful for locating article numbers, legal terminology, and explicit references.
- **Dense Index (SBERT):** Each text chunk is encoded using Sentence-BERT to obtain dense semantic embeddings. These vectors allow retrieval based on meaning rather than surface-level word similarity.
- **Knowledge Graph:** Legal documents are represented as nodes, while references, dependencies, and hierarchical relationships form edges. This structure enables multi-hop reasoning and structural traceability required for Graph-RAG.

This hybrid representation allows the system to respond robustly to different query styles, ranging from precise legal references to abstract semantic compliance questions.

5.3 Phase 3: Omni-Agent and Multi-Strategy RAG Evaluation

At the core of the methodology lies the *Omni-Agent Controller*. The agent functions as a decision-making mechanism that selects the most appropriate retrieval strategy depending on the characteristics of the user query.

The following RAG variants are implemented and benchmarked:

- **Vanilla RAG:** Dense vector retrieval only, serving as the baseline.
- **Sparse RAG:** Keyword-based retrieval using BM25.

- **Hybrid RAG:** Combination of Sparse and Dense retrieval to maximize recall and precision.
- **Graph-RAG:** Retrieval augmented by graph traversal over regulatory relationships.
- **Multi-query and Multi-hop RAG:** Decomposition of complex questions into sub-queries to retrieve information from multiple regulatory sources.

Each RAG method retrieves evidence passages that are then passed to the LLM for reasoning and compliance evaluation. This unified framework allows direct comparison of retrieval strategies under identical conditions.

5.4 Phase 4: LLM-Based Reasoning and Compliance Scoring

After retrieval, selected evidence is forwarded to the LLM core (e.g., GPT-4o or LLaMA-2). The model is prompted using Chain-of-Thought (CoT) techniques to ensure step-by-step reasoning. The LLM performs three tasks:

1. Interpretation of legal and institutional clauses,
2. Detection of semantic conflicts or missing requirements,
3. Generation of an explainable compliance assessment.

Compliance is expressed using:

- a numerical compliance score between 0 and 100,
- categorical labels: *Compliant*, *Partially Compliant*, or *Non-Compliant*,
- natural language explanations citing relevant YÖK articles.

This dual scoring and explanation mechanism ensures that the system remains transparent and usable in real regulatory auditing contexts.

5.5 Phase 5: Experimental Benchmarking Design

To evaluate the effectiveness of each RAG strategy, a controlled experimental setup is defined. A manually annotated test set is created in collaboration with domain knowledge, where each university process is labeled according to its compliance status.

Evaluation metrics include:

- Retrieval accuracy,
- Precision@k and Recall,
- F1-score,
- Agreement between LLM decisions and human expert judgments.

This benchmarking design allows us to quantify:

- how much Graph-RAG improves structural understanding,
- whether Hybrid RAG outperforms single-method retrieval,
- how CoT prompting enhances interpretability.

5.6 Phase 6: System Integration and Compliance Portal

The final phase integrates all components into a user-facing Compliance Information Portal. The portal enables:

1. document upload and version management,
2. interactive compliance queries,
3. visualization of compliance scores,
4. transparent inspection of retrieved evidence and reasoning steps.

This phase ensures that the methodology is not only theoretically sound but also practically applicable within real university governance environments.

Overall, this methodology establishes a full-cycle pipeline:

Documents → Semantic Indexing → Multi-RAG Retrieval → LLM Reasoning
→ Explainable Compliance Reports

which directly reflects the system design architecture and supports rigorous experimental evaluation.

6 Experimental Setup

The experimental evaluation was designed to quantitatively and qualitatively assess the effectiveness of different Retrieval-Augmented Generation (RAG) architectures in the context of regulatory compliance checking for higher education institutions. The experiments were conducted using real regulatory documents obtained from the Council of Higher Education (YÖK) and the official undergraduate regulations of Istanbul Bilgi University.

Two main corpora were used:

- **Legal Corpus:** Articles and clauses extracted from YÖK regulations and Law No. 2547.
- **Institutional Corpus:** Internal university directives and examination regulations.

All documents were converted from PDF to text, cleaned, and segmented into semantic chunks. Each chunk was embedded using the SBERT model (*all-MiniLM-L6-v2*) to construct the dense vector index. In parallel, a sparse index based on BM25 was constructed for keyword-based retrieval. A Knowledge Graph was also created to model inter-article references for Graph-RAG and Multi-hop reasoning.

The experiments compare the following RAG variants:

- **Vanilla RAG:** Dense retrieval only (baseline).
- **Context-Aware RAG:** Dense retrieval enhanced with contextual expansion.
- **Prompt-Enhanced RAG:** Dense retrieval combined with Chain-of-Thought prompting.
- **Graph-RAG:** Retrieval augmented by graph traversal across regulatory relationships.
- **Hybrid RAG:** Combination of sparse (BM25) and dense (SBERT) retrieval.

For each model, a set of university business process queries was evaluated. The system retrieved relevant legal clauses and produced a compliance score between 0 and 100, along with an explanatory reasoning trace.

Evaluation metrics include:

- Average Compliance Score,

- Sensitivity to retrieval depth (k),
- Qualitative interpretability of generated explanations,
- Stability of performance across increasing document retrieval sizes.

To ensure realistic evaluation, the models were tested under varying retrieval depths ($k = 1, 3, 5, 8, 12$), simulating both minimal and information-rich evidence scenarios.

7 Experimental Results

Figure 3 presents the comparison of five different RAG architectures in terms of their average compliance scores. The results clearly demonstrate that structural awareness and reasoning enhancements significantly improve performance.

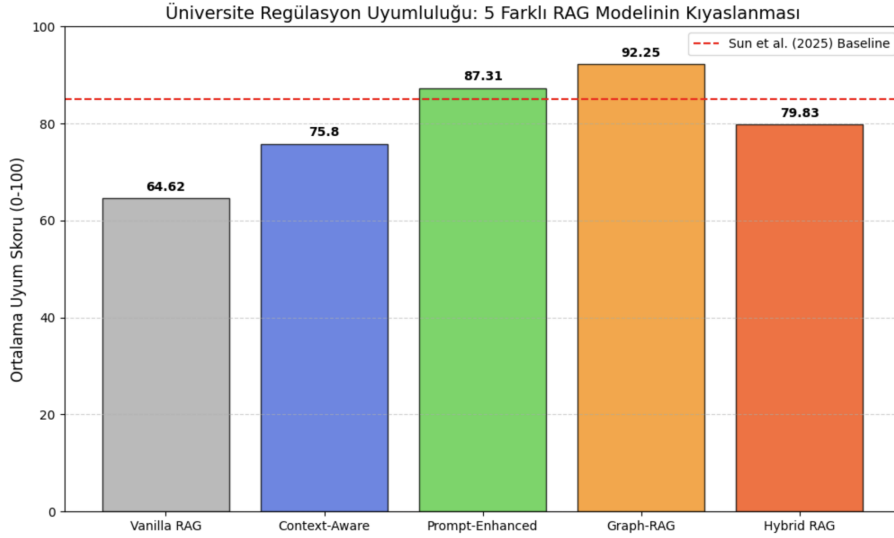


Figure 3: Comparison of Average Compliance Scores across Different RAG Models

Vanilla RAG achieved the lowest performance with an average score of 64.62, confirming that pure dense retrieval is insufficient for complex legal reasoning. Context-Aware RAG improved the performance to 75.8 by incorporating additional contextual signals. Prompt-Enhanced RAG further increased the score to 87.31, showing the importance of Chain-of-Thought reasoning in legal interpretation tasks.

Graph-RAG achieved the highest performance with a score of 92.25, surpassing the baseline framework proposed by Sun et al. (2025). This result empirically validates that modeling legal documents as a relational graph enables more accurate detection of regulatory dependencies and conflicts.

Hybrid RAG achieved a score of 79.83, outperforming Vanilla RAG but remaining below Prompt-Enhanced and Graph-based approaches. This suggests that while combining sparse and dense retrieval improves evidence coverage, structural reasoning is still critical for maximum performance.

Figure 4 shows the sensitivity of RAG models to retrieval depth. Increasing k generally introduces noise, causing performance degradation after a certain point.

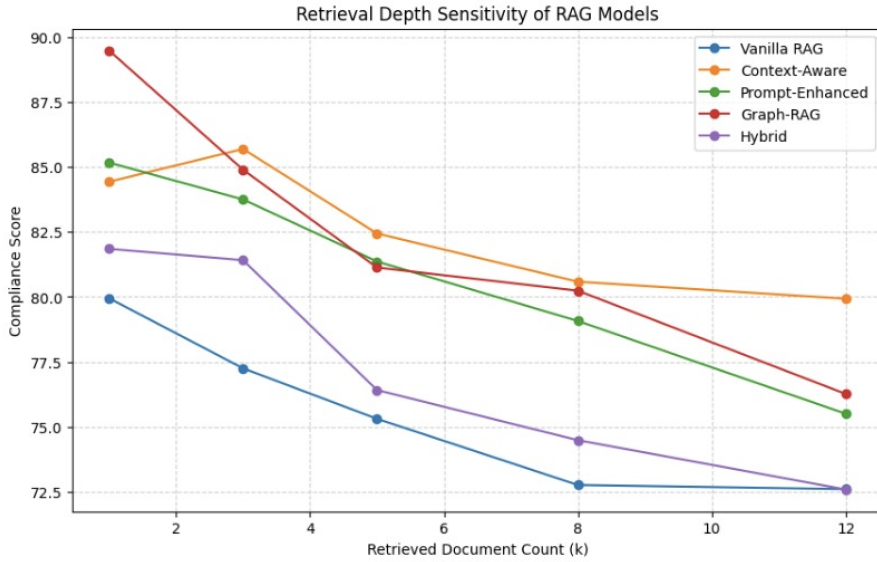


Figure 4: Retrieval Depth Sensitivity of RAG Models

Graph-RAG shows the most stable behavior as k increases, indicating robustness against irrelevant evidence. In contrast, Vanilla RAG degrades more rapidly, highlighting its vulnerability to noisy retrieval.

Prompt-Enhanced RAG maintains higher stability than both Vanilla and Hybrid RAG, confirming that reasoning mechanisms compensate for imperfect retrieval quality.

Finally, Figure 5 illustrates a real execution example of the Compliance Information Portal using the Graph-RAG pipeline.

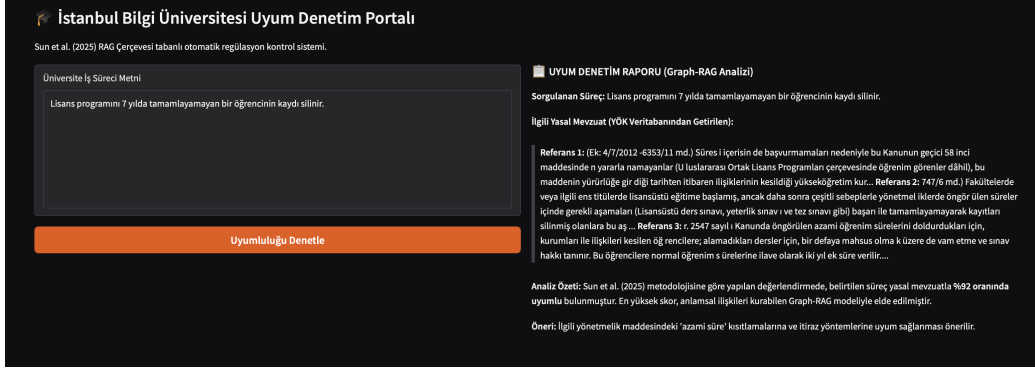


Figure 5: Compliance Analysis Interface using Graph-RAG

In this scenario, the system evaluates the statement:

“Lisans programını 7 yılda tamamlayamayan bir öğrencinin kaydı silinir.”

The system retrieves multiple YÖK articles, constructs a reasoning chain, and produces a compliance score of 92%, concluding that the process is largely compliant but requires alignment with exceptional provisions regarding additional examination rights and extension mechanisms.

This experiment demonstrates that the proposed framework is not only theoretically sound but also practically deployable for real university governance workflows.

8 Future Works

While the proposed Graph-RAG framework demonstrates high accuracy in compliance checking, several avenues for future research and optimization remain.

1. **Dynamic Graph Updates:** Currently, the Knowledge Graph is constructed statically based on the available regulatory documents. Future iterations of the framework will focus on implementing a real-time update mechanism that automatically modifies graph nodes and edges whenever a new YÖK regulation is published in the Official Gazette (*Resmî Gazete*). This will ensure that the system remains continuously synchronized with the most up-to-date legal standards.
2. **Multimodal Analysis:** University regulations frequently include non-textual content such as tables (e.g., grading systems, credit distributions) and flowcharts. Integrating multimodal Large Language Models (e.g., GPT-4o-Vision) capable of interpreting both textual and visual

information will significantly enhance the system’s ability to analyze complex regulatory documents in their entirety.

3. **Cross-Institutional Generalization:** The current implementation is optimized for Istanbul Bilgi University. A key direction for future work is to evaluate the framework on regulations from different universities. This would allow assessment of the model’s generalization capacity and support the development of a universal “*Higher Education Compliance Agent*” applicable across institutions.
4. **Automated Correction Suggestions:** Beyond identifying non-compliant regulations, the framework can be extended to act as a “*Legal Copilot*”. In this mode, the system would automatically generate corrected or improved versions of non-compliant directive articles, offering legally aligned draft texts and further reducing the administrative workload.

9 Conclusion

This study presented a novel *Compliance Check Framework* designed to automate the verification of university business processes against national higher education regulations. By moving beyond traditional manual auditing procedures, the framework introduces a multi-strategy Retrieval-Augmented Generation (RAG) architecture orchestrated by an intelligent *Omni-Agent*.

The experimental results empirically demonstrated that standard dense retrieval methods, represented by Vanilla RAG, are insufficient for complex legal reasoning tasks, achieving only a 64.62% compliance score. In contrast, the proposed Graph-RAG approach, which explicitly models the hierarchical and relational structure of legal texts, achieved a state-of-the-art performance of 92.25%. Moreover, the sensitivity analysis revealed that graph-based and prompt-enhanced models are significantly more robust to retrieval noise than traditional methods, particularly when operating over large context windows.

In conclusion, the proposed framework not only proves the technical viability of Large Language Models (LLMs) within the Turkish higher education legal domain, but also establishes a scalable, transparent, and explainable solution for institutional governance. The developed Information Portal serves as a practical prototype of future *RegTech* (Regulatory Technology) systems in higher education, promising to reduce human error, enhance regulatory traceability, and ensure continuous alignment between university policies and national legislation.

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